

TAMPERE HARMALA, Finland

WMO: 027630

Lat: 61.4656N

Lon: 23.7468E

Elev: 282

StdP: 14.55

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.8	(c) -4.3	(d) -14.5	(e) 2.5	(f) -9.8	(g) -8.5	(h) 3.5	(i) -3.5	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.2	81.6	65.4	78.0	63.3	74.5	61.9	68.3	76.2	66.1	74.3	63.9	71.8	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	94.9	72.0	63.0	87.1	69.5	60.8	80.3	67.1	32.7	76.2	30.9	74.4	29.3	72.0	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(e) -12.3	(f) 84.1	(g) 7.8	(h) 4.3	(i) -17.9	(j) 87.2	(k) -22.4	(l) 89.8	(m) -26.8	(n) 92.2	(o) -32.4	(p) 95.3		
(4) N/A	(5) N/A	(6) N/A	DB	-12.7	69.7	7.8	3.3	-18.2	72.1	-22.8	74.0	-27.1	75.9	-32.7	78.3	(4) (5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.3	19.5	23.1	28.6	39.1	50.9	57.4	63.3	60.0	51.6	40.3	34.1	27.0		
	DBStd	16.53	11.59	11.36	8.38	6.30	7.56	6.23	5.85	5.31	5.81	6.45	7.61	10.06		
	HDD50	4326	945	753	663	331	81	9	0	1	50	305	476	712		
	HDD65	8716	1410	1173	1128	776	441	241	105	173	401	765	926	1177		
	CDD50	1168	0	0	0	5	108	231	411	309	99	5	0	0		
	CDD65	83	0	0	0	0	3	12	51	17	0	0	0	0		
	CDH74	842	0	0	0	0	70	168	487	117	0	0	0	0		
	CDH80	158	0	0	0	0	5	31	102	20	0	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	22.3	1.4	1.0	1.1	1.3	1.4	2.0	2.8	2.9	2.4	2.2	2.1	1.7		
	PrecMax	34.0	2.9	2.6	2.4	3.5	3.5	6.7	5.2	7.2	4.4	5.1	3.8	3.6		
	PrecMin	15.8	0.5	0.2	0.0	0.0	0.3	0.6	0.8	0.9	0.2	0.5	0.4	0.3		
	PrecStd	3.8	0.6	0.6	0.6	0.8	0.7	1.3	1.3	1.5	1.0	1.1	0.9	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.3	42.5	51.0	67.9	80.0	84.0	86.5	83.2	70.8	57.5	50.6	47.1		
		MCWB	38.2	37.8	41.1	50.8	59.6	65.0	67.2	67.5	61.7	53.2	48.7	46.1		
	2%	DB	36.9	38.7	45.1	61.0	75.6	79.0	83.1	76.9	66.6	54.0	48.1	41.7		
		MCWB	35.8	36.5	38.6	47.3	58.0	62.1	67.0	63.5	59.0	51.1	47.0	40.3		
	5%	DB	35.0	36.9	41.7	55.9	71.3	74.5	79.5	73.5	63.4	52.4	45.3	39.3		
		MCWB	34.1	35.6	36.9	44.9	56.5	60.4	65.3	61.8	57.2	50.1	44.3	38.3		
	10%	DB	33.5	35.0	38.8	51.3	66.8	70.8	75.9	70.3	61.0	50.3	43.2	37.4		
		MCWB	32.6	34.1	35.3	42.5	54.0	58.9	64.2	60.6	56.5	48.2	42.3	36.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	38.6	42.8	52.8	63.0	68.3	71.2	69.7	62.6	54.6	49.2	46.3		
		MCDB	39.0	40.2	47.7	63.5	72.4	77.3	81.1	77.8	68.4	56.8	50.2	47.0		
	2%	WB	36.2	37.0	39.6	48.8	60.3	64.7	69.2	66.4	60.5	52.3	46.9	40.4		
		MCDB	36.8	38.1	43.9	58.7	72.2	74.6	77.9	73.2	64.7	53.6	48.1	41.3		
	5%	WB	34.2	35.7	37.7	45.9	57.8	62.3	67.5	64.2	58.8	50.5	44.4	38.4		
		MCDB	34.8	36.7	40.7	53.3	68.4	71.9	75.3	70.4	62.2	51.8	45.1	39.1		
	10%	WB	32.8	34.2	35.9	43.6	55.1	60.0	65.5	62.3	57.0	48.2	42.4	36.5		
		MCDB	33.5	35.0	38.3	50.3	64.9	68.4	74.0	68.0	59.9	50.1	43.1	37.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	9.8	14.0	17.2	19.7	17.7	17.2	16.6	13.9	10.9	6.9	7.7		
		MCDBR	6.4	9.0	16.2	27.1	28.7	25.4	22.6	21.6	18.1	12.8	7.8	7.9		
	5% WB	MCWBR	6.3	7.6	10.8	15.1	13.1	11.6	9.4	10.6	11.3	9.9	7.3	7.5		
		MCWBR	6.3	8.3	14.7	22.9	24.2	20.9	19.2	17.8	15.2	10.9	7.6	7.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.251	0.295	0.322	0.356	0.353	0.349	0.374	0.360	0.336	0.321	0.274	0.357		
		taud	2.085	2.166	2.252	2.372	2.444	2.477	2.426	2.456	2.519	2.411	2.184	2.597		
	Ebn at Noon	Ebn at Noon	143	211	250	265	276	279	267	262	249	206	134	95		
		Edh at Noon	12	22	29	31	31	31	32	29	23	18	11	8		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	69	248	715	1167	1588	1688	1596	1249	742	314	82	38		
	RadStd	RadStd	13	56	89	132	170	130	161	139	95	60	16	4		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (59 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	-241	N/A	N/A	N/A

Nomenclature: See separate page